Auteur: Odia Kibor
Studierichting: Informatica
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$A(z) = z^4 - 5z^3 + 9z^2 - 15z + 18$

$A(2) = 0$

$$\begin{array}{c|ccccc} & 1 & -5 & 9 & -15 & 18 \\ 2 & & 2 & -6 & 6 & -18 \\ \hline & 1 & -3 & 3 & -9 & 0 \end{array}$$

$\Rightarrow A(z) = (z-2)(z^3 - 3z^2 + 3z - 9)$

$A(3) = 0$

$$\begin{array}{c|cccc} & 1 & -3 & 3 & -9 \\ 3 & & 3 & 0 & 9 \\ \hline & 1 & 0 & 3 & 0 \end{array}$$

$\Rightarrow A(z) = (z-2)(z-3)(z^2+3)$

$z^2 + 3 = 0 \Leftrightarrow z^2 = -3 \Leftrightarrow z = \pm\sqrt{3}i$

$\Rightarrow A(z) = (z-2)(z-3)(z+\sqrt{3}i)(z-\sqrt{3}i)$

nulpunten: $2, 3, \sqrt{3}i, -\sqrt{3}i$

References